

Linear Relations

First Differences · Direct & Partial Variation · Slope · Intercepts · Equations of Lines · Standard Form

MTH1W · Homework — Independent Practice

Name: _____

Date: _____

Goal: Practise first differences, variation, slope, intercepts, equations and standard form independently.

Show your full working in the space provided under each question. Levels: L1 fluency, L2 application, L3 thinking. Full worked solutions are in the Answer Key.

Level 1 — Fluency

1. The y-values in a table are 6, 10, 14, 18. Find the first differences and state whether the relation is linear. **[First differences]**
2. Is $y = 7x + 1$ direct or partial variation? Give a reason. **[Direct or partial]**
3. Find the slope of the line through (2, 4) and (5, 13). **[Slope]**
4. State the y-intercept of the line $y = 5x - 8$. **[y-intercept]**

5. Find the x-intercept of the line $y = 2x - 10$. **[x-intercept]**
6. Write $y = 4x + 1$ in the standard form $Ax + By + C = 0$. **[Standard form]**

Level 2 — Application

7. The y-values in a table are 3, 6, 11, 18. Find the first differences and decide whether the relation is linear. **[First differences]**
8. A direct variation passes through (5, 15). Find the constant k and write the equation. **[Direct variation]**

9. Find the slope of the line through $(-3, 8)$ and $(1, -4)$. **[Slope]**
10. Find the x-intercept and the y-intercept of $3x + 4y = 24$. **[Both intercepts]**
11. Find the equation of the line with slope 2 that passes through $(3, 5)$. **[Equation of a line]**
12. Write $y = (2/3)x - 4$ in standard form with integer coefficients. **[Standard form]**

Level 3 — Thinking

13. A linear table gives $y = 1, ?, 13, 19$ for $x = 0, 1, 2, 3$. Find the missing value and explain. **[Missing table value]**
14. A gym charges a \$20 joining fee plus \$15 per month: $C = 15m + 20$. Find the cost for 6 months and state the fixed value. **[Partial variation model]**
15. The line through $(2, 5)$ and $(k, 17)$ has slope 4. Find k . **[Unknown coordinate]**

16. A line has x-intercept -4 and y-intercept 8. Write its equation. **[Equation from intercepts]**
17. Find the equation of the line through (-1, 2) and (3, 14). **[Line through two points]**
18. A scatter plot of ice-cream sales versus daily temperature shows points rising to the right in a tight band. Describe the correlation and what it suggests. **[Correlation reasoning]**